

Scope Of the project

Project Title:

CrewOpsAI

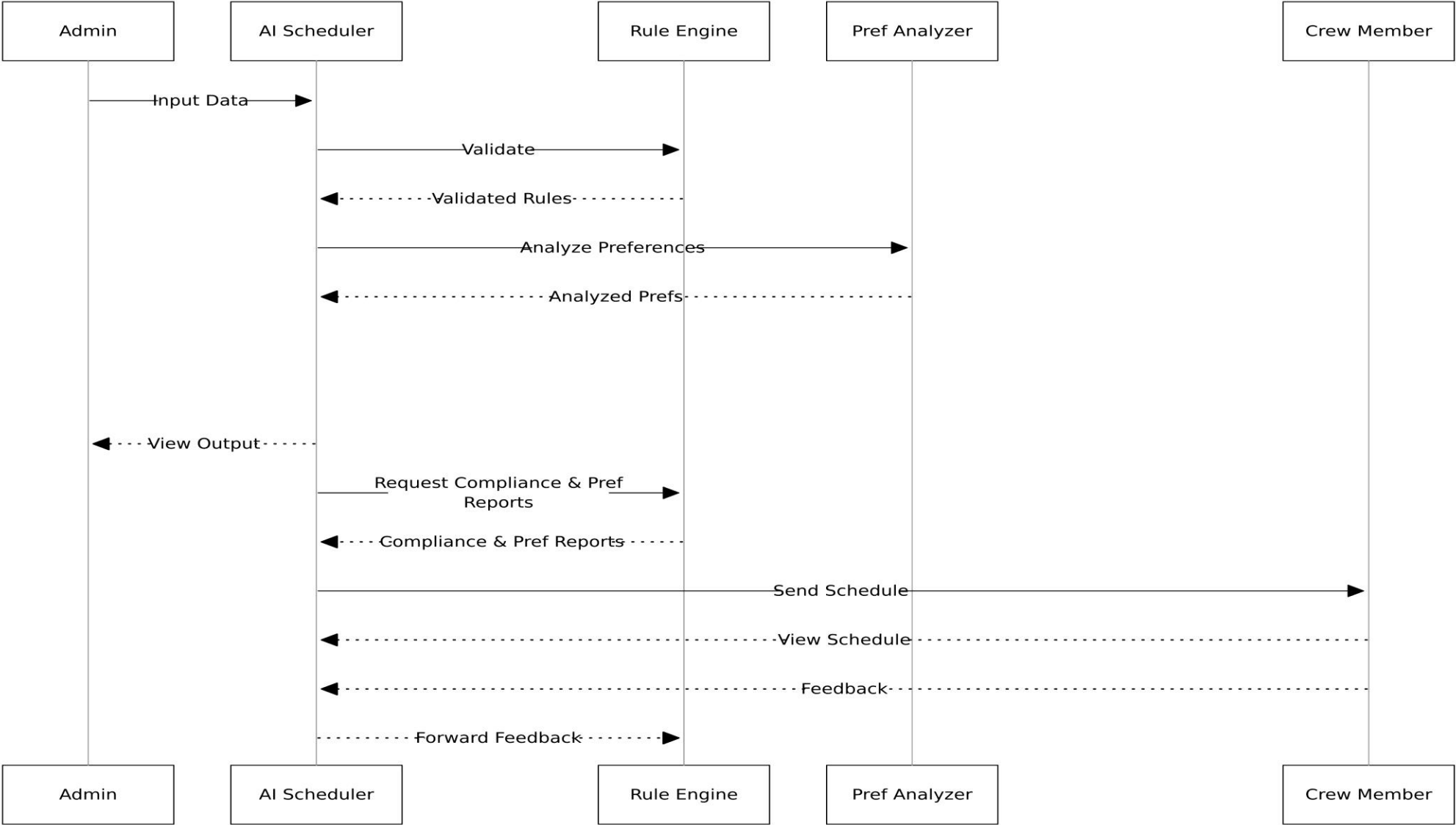
Project Objectives:

- Automates crew assignments while ensuring compliance with regulations, rest requirements, and employee preferences.
- Reduces operational costs and improves employee satisfaction by minimizing scheduling conflicts.

Deliverables:

- The core deliverables include single software application for Admins and Crew Members which comprises of modules for AI Scheduling, Rule Engine logic, Preference Analysis, and Compliance Reporting, along with the necessary software layers to integrate with databases and LLMs.

Design Diagram



Design Description:

Core Components:

- *Admin:*
 - Inputs data and rules into the system.
 - Uses the Input Data Database and Rules Database for validation.
 - Requests compliance and preference reports.
- *AI Scheduler:*
 - Uses the results from Preferences Analyzer Module and Rule Engine Module to create Schedules.
 - Works with the LLM to process the results of references Analyzer Module and Rule Engine Module.
 - Sends schedules to Crew Members.
- *Rule Engine:*
 - Validates rules using the Rules Database.
 - Assisted by the LLM for Rule Validation for complex or natural language rules.
 - Works with the AI Scheduler.
- *Preference Analyzer:*
 - Analyzes crew preferences using the Preferences Database.
 - Enhanced by the LLM for Preference Analysis to understand natural language preference inputs.
 - Provides insights to the AI Scheduler.
- *Crew Member:*
 - Views schedules sent by the AI Scheduler.
 - Provides feedback, which is stored in the Feedback Database.

Design Description:

Supporting Components:

- *Databases:*
 - Input Data Database: Stores raw input data.
 - Rules Database: Stores predefined rules.
 - Preferences Database: Stores crew member preferences.
 - Feedback Database: Stores crew member feedback.
- *LLMs (Large Language Models):*
 - LLM for Preference Analysis: Interprets natural language preferences.
 - LLM for Rule Validation: Helps validate complex/natural language rules.
 - LLM for Feedback Analysis: Processes feedback for insights.

Workflow:

The Admin initiates by inputting data and rules, which are validated against respective databases aided by an LLM for rules. The AI Scheduler, using preferences (analyzed by the Preference Analyzer and its LLM) and rules (validated by the Rule Engine and its LLM), generates and sends schedules to Crew Members. Crew Members view schedules and provide feedback, which is stored and analyzed (using an LLM) to improve the system continuously.

Test Cases

Positive Test cases:

#TC1

- Verify that an Admin can successfully input valid crew member data.
- Verify that an Admin can successfully input valid scheduling rules.
- Verify that the system accepts and stores valid data in the Input Data and Rules Databases.

#TC2

- Verify that the Rule Engine correctly validates a set of compatible, predefined rules from the Rules Database.
- Verify that the LLM for Rule Validation successfully interprets and validates a rule written in clear natural language.

#TC3

- Verify that the Preference Analyzer correctly processes structured preference data from the Preferences Database.
- Verify that the LLM for Preference Analysis successfully interprets and categorizes a crew member's preference submitted in natural language (e.g., "I prefer to work weekends").

#TC4

- Verify that the AI Scheduler successfully generates a valid schedule when all rules and preferences are compatible and satisfiable.
- Verify that the generated schedule correctly reflects a specific high-priority preference from the Preferences Database.
- Verify that the generated schedule adheres to all validated rules from the Rules Database.

#TC5

- Verify that a Crew Member can successfully view their assigned schedule via their interface.
- Verify that a Crew Member can successfully submit feedback, and it is stored in the Feedback Database.

Test Cases

Negative Test cases:

#TC1

- Verify that the system rejects attempts to input data with missing required fields.

#TC2

- Verify that the Rule Engine flags conflicting rules within the Rules Database.
- Verify that the LLM for Rule Validation flags ambiguous or incomprehensible natural language rules.

#TC3

- Verify that the system handles invalid or improperly formatted preference data gracefully (e.g., ignores it or flags for review).
- Verify that the LLM for Preference Analysis flags or asks for clarification for vague or contradictory natural language preferences.

#TC4

- Verify that the AI Scheduler fails to generate a schedule (or flags conflicts) when rules and preferences are inherently contradictory and cannot be satisfied.

#TC5

- Verify system behavior (e.g., fallback to basic processing, error message) if an LLM service (Preference, Rule, or Feedback Analysis) is unresponsive or returns an error.

Tools and Code details

Third party tools Details:

Tools name	Open source/Licensed	URL	Purpose
Chroma	Open Source	https://github.com/chroma-core/chroma	Storage of embeddings and metadata
AWS	Licensed	aws.amazon.com	Cloud Hosting and Infrastructure

Technologies used to develop in this project

Technology name	Version
Python	3.13.3
LangChain, LangGraph, LangSmith	V0.3
Streamlit	1.44

Code details

Code Repo Details

<https://github.com/shub8968/CrewOpsAI>